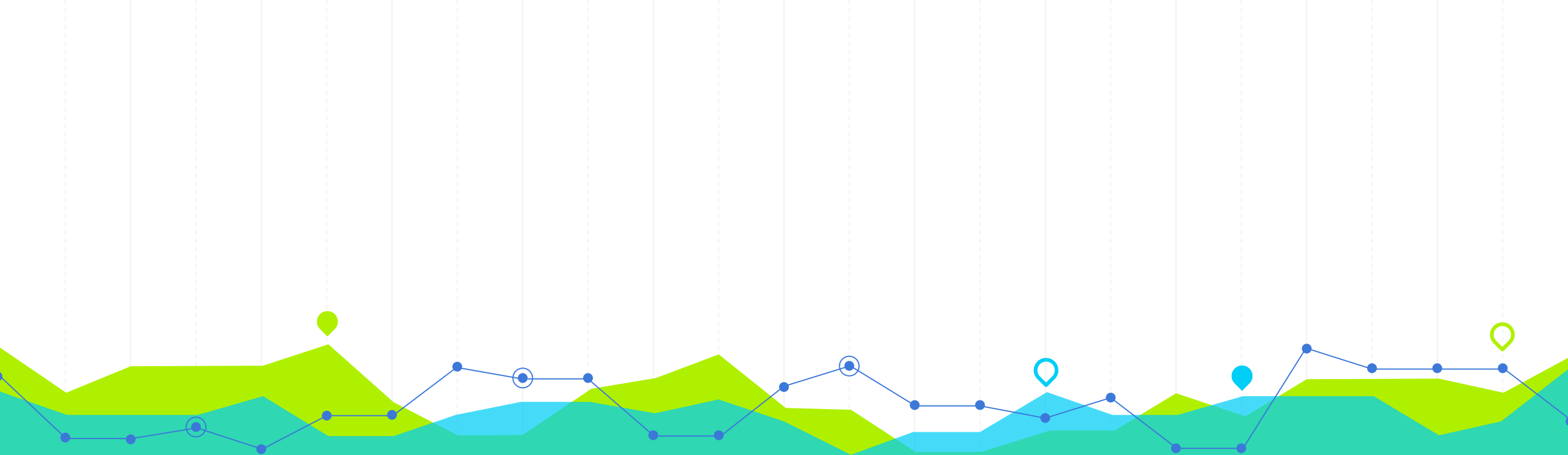




Multivariate Probability Distribution

Marginal and Conditional Probability Distributions

Introduction

The distinct values assumed by a discrete random variable represent mutually exclusive events.

Similarly, for all distinct pairs of values y_1, y_2 , the bivariate events $(Y_1 = y_1, Y_2 = y_2)$, represented by (y_1, y_2) , are mutually exclusive events.

It follows that the univariate event $(Y_1 = y_1)$ is the union of bivariate events of the type $(Y_1 = y_1, Y_2 = y_2)$, with the union being taken over all possible values for y_2 .



For example, reconsider the die-tossing experiment of previous lecture , where

Y_1 = number of dots on the upper face of die 1,

Y_2 = number of dots on the upper face of die 2 . Then,

$$\begin{aligned} P(Y_1 = 1) &= p(1, 1) + p(1, 2) + p(1, 3) + \cdots + p(1, 6) \\ &= 1/36 + 1/36 + 1/36 + \cdots + 1/36 = 6/36 = 1/6 \end{aligned}$$

$$P(Y_1 = 2) = p(2, 1) + p(2, 2) + p(2, 3) + \cdots + p(2, 6) = 1/6$$

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$$P(Y_1 = 6) = p(6, 1) + p(6, 2) + p(6, 3) + \cdots + p(6, 6) = 1/6.$$

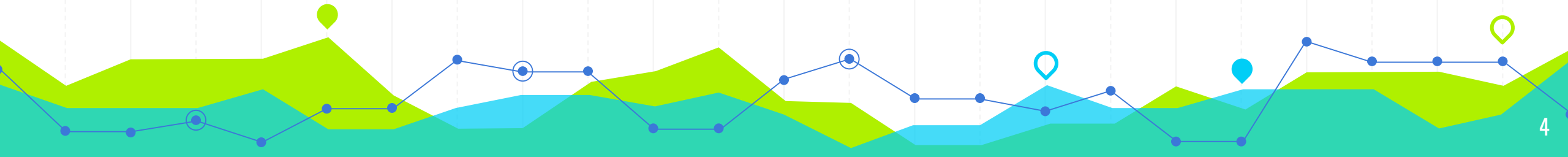
Expressed in summation notation, probabilities about the variable Y_1 alone are

$$P(Y_1 = y_1) = p_1(y_1) = \sum_{y_2=1}^6 p(y_1, y_2).$$

Similarly, probabilities corresponding to values of the variable Y_2 alone are given by

$$p_2(y_2) = P(Y_2 = y_2) = \sum_{y_1=1}^6 p(y_1, y_2).$$

Summation in the discrete case corresponds to integration in the continuous case, which leads us to the following definition.



DEFINITION 4

- Let Y_1 and Y_2 be jointly discrete random variables with probability function $p(y_1, y_2)$. Then the marginal probability functions of Y_1 and Y_2 , respectively, are given by

$$p_1(y_1) = \sum_{\text{all } y_2} p(y_1, y_2) \quad \text{and} \quad p_2(y_2) = \sum_{\text{all } y_1} p(y_1, y_2).$$

- Let Y_1 and Y_2 be jointly continuous random variables with joint density function $f(y_1, y_2)$. Then the marginal density functions of Y_1 and Y_2 , respectively, are given by

$$f_1(y_1) = \int_{-\infty}^{\infty} f(y_1, y_2) dy_2 \quad \text{and} \quad f_2(y_2) = \int_{-\infty}^{\infty} f(y_1, y_2) dy_1.$$

Example 4

From a group of three Republicans, two Democrats, and one independent, a committee of two people is to be randomly selected. Let Y_1 denote the number of Republicans and Y_2 denote the number of Democrats on the committee. Find the joint probability function of Y_1 and Y_2 and then find the marginal probability function of Y_1 .

Example 5

Let

$$f(y_1, y_2) = \begin{cases} 2y_1, & 0 \leq y_1 \leq 1, 0 \leq y_2 \leq 1, \\ 0, & \text{elsewhere.} \end{cases}$$

Sketch $f(y_1, y_2)$ and find the marginal density functions for Y_1 and Y_2 .



Conditional Distributions

Discrete Case

The multiplicative law gives the probability of the intersection $A \cap B$ as

$$P(A \cap B) = P(A)P(B|A),$$

where $P(A)$ is the unconditional probability of A and $P(B|A)$ is the probability of B given that A has occurred.



Now consider the **intersection of the two numerical events**,
($Y_1 = y_1$) and ($Y_2 = y_2$), represented by the bivariate event (y_1, y_2).

It follows directly from the multiplicative law of probability that the bivariate probability for the intersection (y_1, y_2) is

$$p(y_1, y_2) = p_1(y_1)p(y_2 | y_1) = p_2(y_2)p(y_1 | y_2).$$

The probabilities $p_1(y_1)$ and $p_2(y_2)$ are associated with the univariate probability distributions for Y_1 and Y_2 individually.

$p(y_1 | y_2)$ is the probability that the random variable Y_1 equals y_1 , given that Y_2 takes on the value y_2 .



DEFINITION 5

If Y_1 and Y_2 are jointly discrete random variables with joint probability function $p(y_1, y_2)$ and marginal probability functions $p_1(y_1)$ and $p_2(y_2)$, respectively, then the conditional discrete probability function of Y_1 given Y_2 is

$$\begin{aligned} p(y_1|y_2) &= P(Y_1 = y_1 | Y_2 = y_2) \\ &= \frac{P(Y_1 = y_1, Y_2 = y_2)}{P(Y_2 = y_2)} = \frac{p(y_1, y_2)}{p_2(y_2)}, \end{aligned}$$

provided that $p_2(y_2) > 0$.

Example 6

Refer to Example 4 and find the conditional distribution of Y_1 given that $Y_2 = 1$. That is, given that one of the two people on the committee is a Democrat, find the conditional distribution for the number of Republicans selected for the committee.



Continuous Case

If Y_1 and Y_2 are continuous, $P(Y_1 = y_1 | Y_2 = y_2)$ cannot be defined as in the discrete case because both $(Y_1 = y_1)$ and $(Y_2 = y_2)$ are events with zero probability.

Assuming that Y_1 and Y_2 are jointly continuous with density function $f(y_1, y_2)$, we might be interested in a probability of the form

$$P(Y_1 \leq y_1 | Y_2 = y_2) = F(y_1 | y_2),$$

which, as a function of y_1 for a fixed y_2 , is called the **conditional distribution function** of Y_1 , given $Y_2 = y_2$.

DEFINITION 6

If Y_1 and Y_2 are jointly continuous random variables with joint density function $f(y_1, y_2)$, then the conditional distribution function of Y_1 given

$Y_2 = y_2$ is

$$F(y_1|y_2) = P(Y_1 \leq y_1 | Y_2 = y_2).$$

Notice that $F(y_1|y_2)$ is a function of y_1 for a fixed value of y_2 .

If we could take $F(y_1|y_2)$, multiply by $P(Y_2 = y_2)$ for each possible value of Y_2 , and sum all the resulting probabilities, we would obtain $F(y_1)$.

This is not possible because the number of values for y_2 is uncountable and all probabilities $P(Y_2 = y_2)$ are zero.

But we can do something analogous by multiplying by $f_2(y_2)$ and then integrating to obtain

$$F(y_1) = \int_{-\infty}^{\infty} F(y_1|y_2) f_2(y_2) dy_2.$$



The quantity $f_2(y_2) dy_2$ can be thought of as the approximate probability that Y_2 takes on a value in a small interval about y_2 , and the integral is a generalized sum.

Now from previous considerations, we know that

$$\begin{aligned} F(y_1) &= \int_{-\infty}^{y_1} f_1(t_1) dt_1 = \int_{-\infty}^{y_1} \left[\int_{-\infty}^{\infty} f(t_1, y_2) dy_2 \right] dt_1 \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{y_1} f(t_1, y_2) dt_1 dy_2. \end{aligned}$$

From these two expressions for $F(y_1)$, we must have

$$F(y_1|y_2)f_2(y_2) = \int_{-\infty}^{y_1} f(t_1, y_2) dt_1$$

or

$$F(y_1|y_2) = \int_{-\infty}^{y_1} \frac{f(t_1, y_2)}{f_2(y_2)} dt_1.$$

We will call the integrand of this expression the conditional density function of Y_1 given $Y_2 = y_2$, and we will denote it by $f(y_1|y_2)$.



DEFINITION 7

Let Y_1 and Y_2 be jointly continuous random variables with joint density $f(y_1, y_2)$ and marginal densities $f_1(y_1)$ and $f_2(y_2)$, respectively. For any y_2 such that $f_2(y_2) > 0$, the conditional density of Y_1 given $Y_2 = y_2$ is given by

$$f(y_1|y_2) = f(y_1, y_2) / f_2(y_2)$$

and, for any y_1 such that $f_1(y_1) > 0$, the conditional density of Y_2 given $Y_1 = y_1$ is given by

$$f(y_2|y_1) = f(y_1, y_2) / f_1(y_1)$$

Note that the conditional density $f(y_1|y_2)$ is undefined for all y_2 such that $f_2(y_2) = 0$. Similarly, $f(y_2|y_1)$ is undefined if y_1 is such that $f_1(y_1) = 0$.

Example 7

A soft-drink machine has a random amount Y_2 in supply at the beginning of a given day and dispenses a random amount Y_1 during the day (with measurements in gallons). It is not resupplied during the day, and hence $Y_1 \leq Y_2$. It has been observed that Y_1 and Y_2 have a joint density given by

$$f(y_1, y_2) = \begin{cases} 1/2, & 0 \leq y_1 \leq y_2 \leq 2, \\ 0 & \text{elsewhere.} \end{cases}$$

That is, the points (y_1, y_2) are uniformly distributed over the triangle with the given boundaries. Find the conditional density of Y_1 given $Y_2 = y_2$. Evaluate the probability that less than 1/2 gallon will be sold, given that the machine contains 1.5 gallons at the start of the day.

Q1

In Exercise Q1(Bivariate and multivariate distribution), we determined that the joint distribution of Y_1 , the number of contracts awarded to firm A, and Y_2 , the number of contracts awarded to firm B, is given by the entries in the following table.

y_2	y_1		
	0	1	2
0	$1/9$	$2/9$	$1/9$
1	$2/9$	$2/9$	0
2	$1/9$	0	0

- Find the marginal probability distribution of Y_1 .
- Y_1 has a binomial distribution with $n = 2$ and $p = 1/3$. Is there any conflict between this result and the answer you provided in part (a)?

Q2

You were given the following joint probability function for

$$Y_1 = \begin{cases} 0, & \text{if child survived,} \\ 1, & \text{if not,} \end{cases} \quad \text{and} \quad Y_2 = \begin{cases} 0, & \text{if no belt used,} \\ 1, & \text{if adult belt used,} \\ 2, & \text{if car-seat belt used.} \end{cases}$$

y_2	y_1		Total
	0	1	
0	.38	.17	.55
1	.14	.02	.16
2	.24	.05	.29
Total	.76	.24	1.00

- Give the marginal probability functions for Y_1 and Y_2
- Give the conditional probability function for Y_2 given $Y_1 = 0$.
- What is the probability that a child survived given that he or she was in a car-seat belt?

Q3

Let Y_1 and Y_2 have joint density function first encountered in previous exercise:

$$f(y_1, y_2) = \begin{cases} e^{-(y_1+y_2)}, & y_1 > 0, y_2 > 0, \\ 0, & \text{elsewhere.} \end{cases}$$

- a) Find the marginal density functions for Y_1 and Y_2 . Identify these densities as one of those studied in Chapter 4.
- b) What is $P(1 < Y_1 < 2.5)$? $P(1 < Y_2 < 2.5)$?
- c) For what values of y_2 is the conditional density $f(y_1|y_2)$ defined?
- d) For any $y_2 > 0$, what is the conditional density function of Y_1 given that $Y_2 = y_2$?

- e) For any $y_1 > 0$, what is the conditional density function of Y_2 given that $Y_1 = y_1$?
- f) For any $y_2 > 0$, how does the conditional density function $f(y_1|y_2)$ that you obtained in part (d) compare to the marginal density function $f_1(y_1)$ found in part (a)?
- g) What does your answer to part (f) imply about marginal and conditional probabilities that Y_1 falls in any interval?



Q4

Suppose that the random variables Y_1 and Y_2 have joint probability density function, $f(y_1, y_2)$, given by

$$f(y_1, y_2) = \begin{cases} 6y_1^2 y_2, & 0 \leq y_1 \leq y_2, y_1 + y_2 \leq 2, \\ 0, & \text{elsewhere.} \end{cases}$$

- a Show that the marginal density of Y_1 is a beta density with $\alpha = 3$ and $\beta = 2$.
- b Derive the marginal density of Y_2 .
- c Derive the conditional density of Y_2 given $Y_1 = y_1$.
- d Find $P(Y_2 < 1.1 | Y_1 = .60)$.

Q5

Let Y_1 denote the weight (in tons) of a bulk item stocked by a supplier at the beginning of a week and suppose that Y_1 has a uniform distribution over the interval $0 \leq y_1 \leq 1$. Let Y_2 denote the amount (by weight) of this item sold by the supplier during the week and suppose that Y_2 has a uniform distribution over the interval $0 \leq y_2 \leq y_1$, where y_1 is a specific value of Y_1 .

- a) Find the joint density function for Y_1 and Y_2 .
- b) If the supplier stocks a half-ton of the item, what is the probability that she sells more than a quarter-ton?
- c) If it is known that the supplier sold a quarter-ton of the item, what is the probability that she had stocked more than a half-ton?